

CPRE 4880 - Section A1

MP - 4

Cameron Gilbertson, Matthew Smith, Jessica Reiff, Collin Van Weelden

09 April 2026

Dr. Phillip Jones

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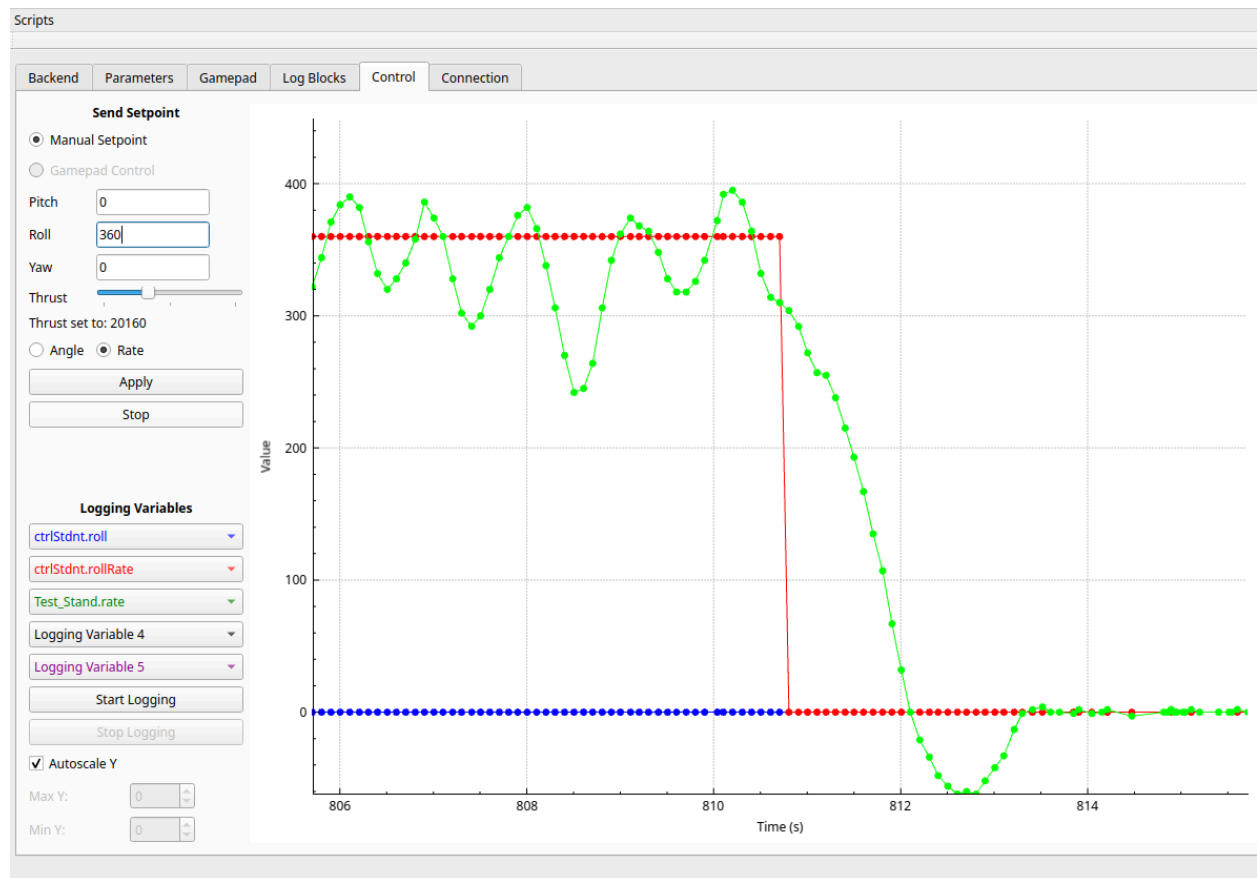
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In your write-up, briefly explain how this Make process was configured to appropriately use a cross-compiler targeting the ARM architecture.....	3
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Part 1 - Relevant Parameters

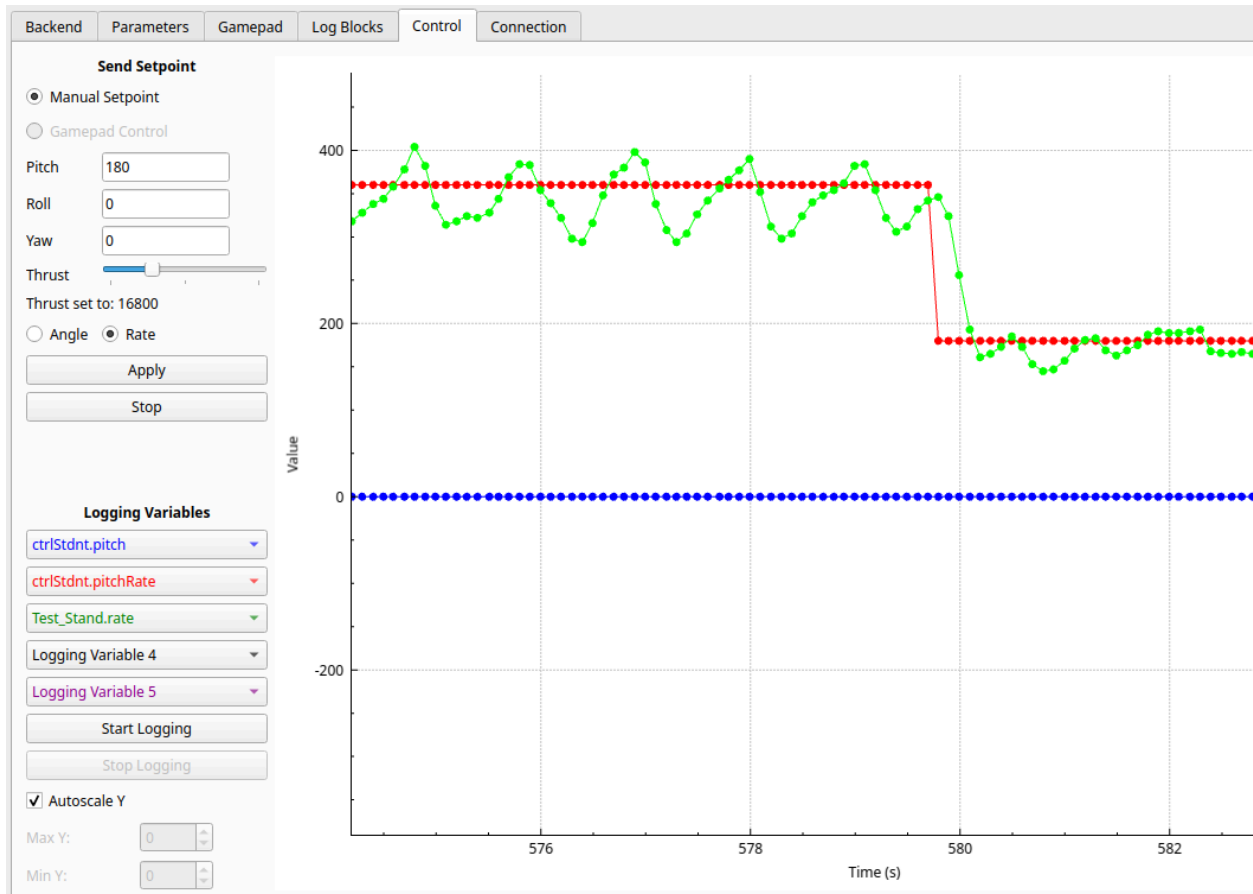
Note: Set thrust to 15,000 - 20,000

Rate	Kp	Ki	Kd
Yaw	100	0	0
Roll	10	0	1
Pitch	50	0	0

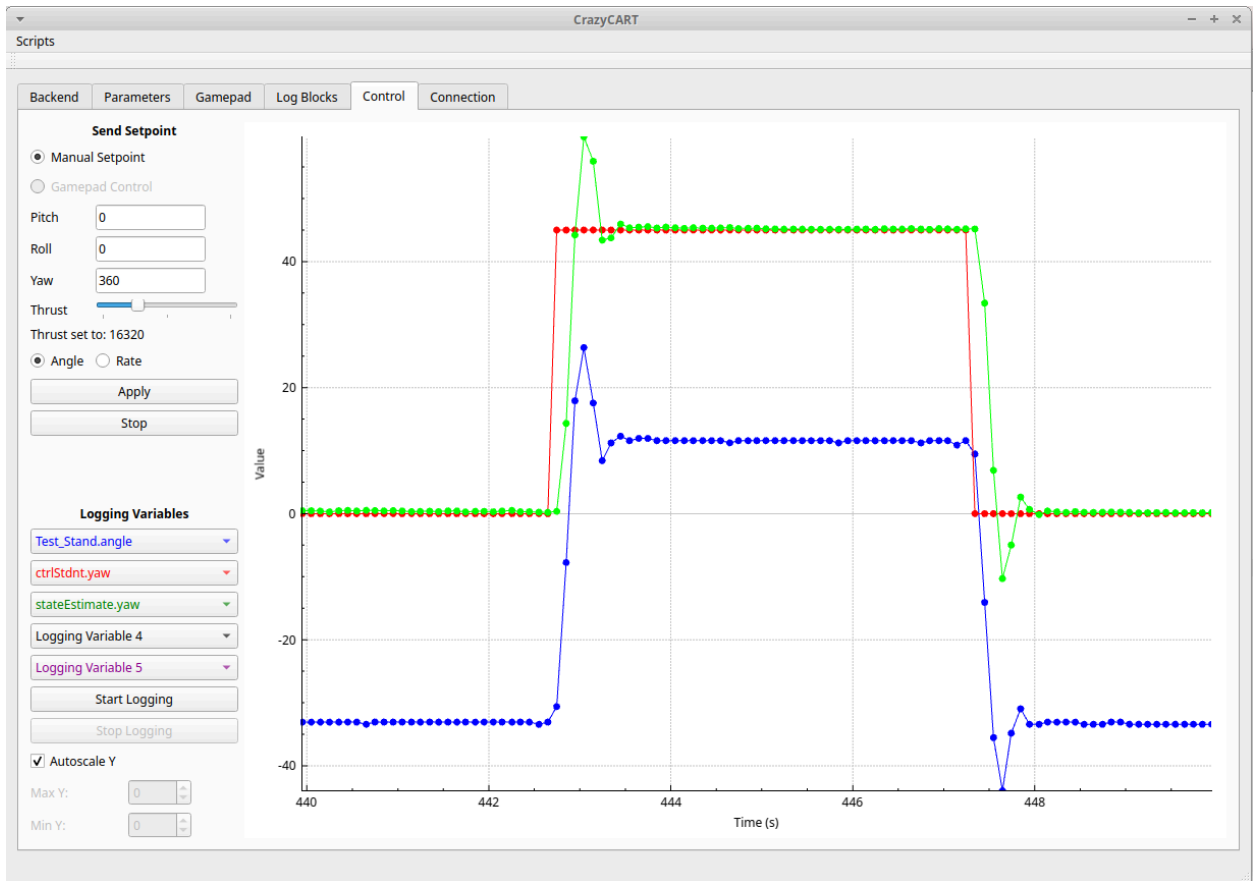
Attitude	Kp	Ki	Kd
Yaw	50	0	1
Pitch	25	0	1
Roll	25	0	0



^^^ For Roll_Rate, our final params being $k_p=10$, $k_d=1$, and $k_i=0$.



^^^ For Roll_Rate, our final params being $k_p=10$, $k_d=1$, and $k_i=0$.



Setpoints

Roll SP: 10.0

Pitch SP: 0.0

Yaw SP: 0.0

Send

☒ Attitude Mode

☐ Rate Mode

Thrust

35000

Input Select

Select File

No File Selected

Model Input

Log File Input

Graph Data Selection

Controls

Build and Run

Stop

Reset

Delete Old Logs

☐ E-Stop Mode

☒ Student Mode

Playback Settings

1.0x (Real-time)

Pause

Camera View

Free Orbit

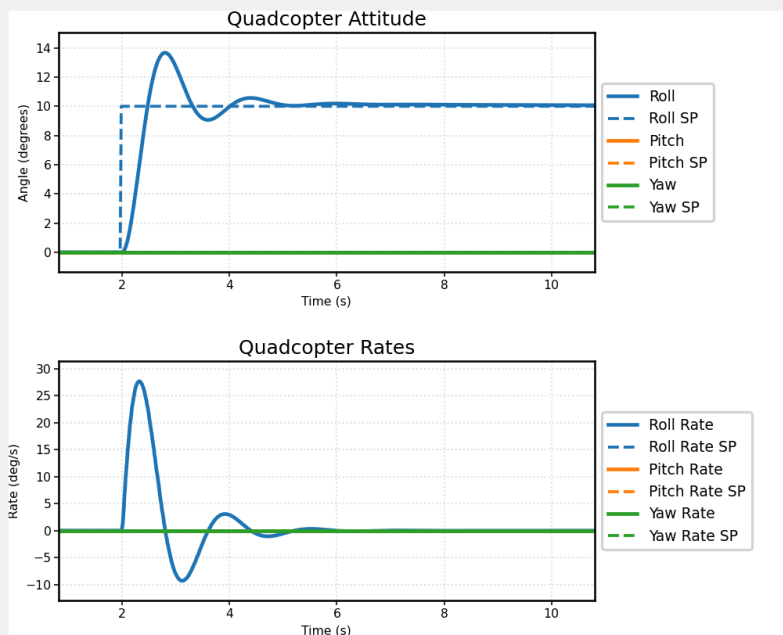
Live Data

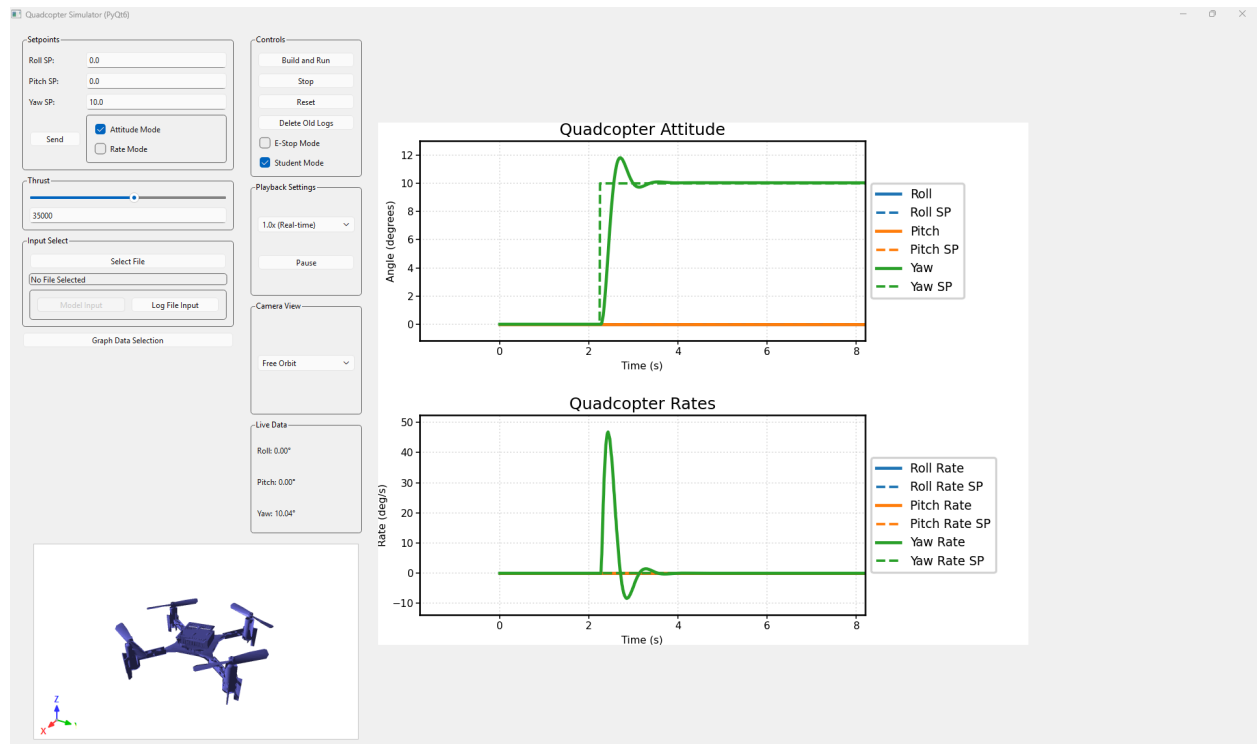
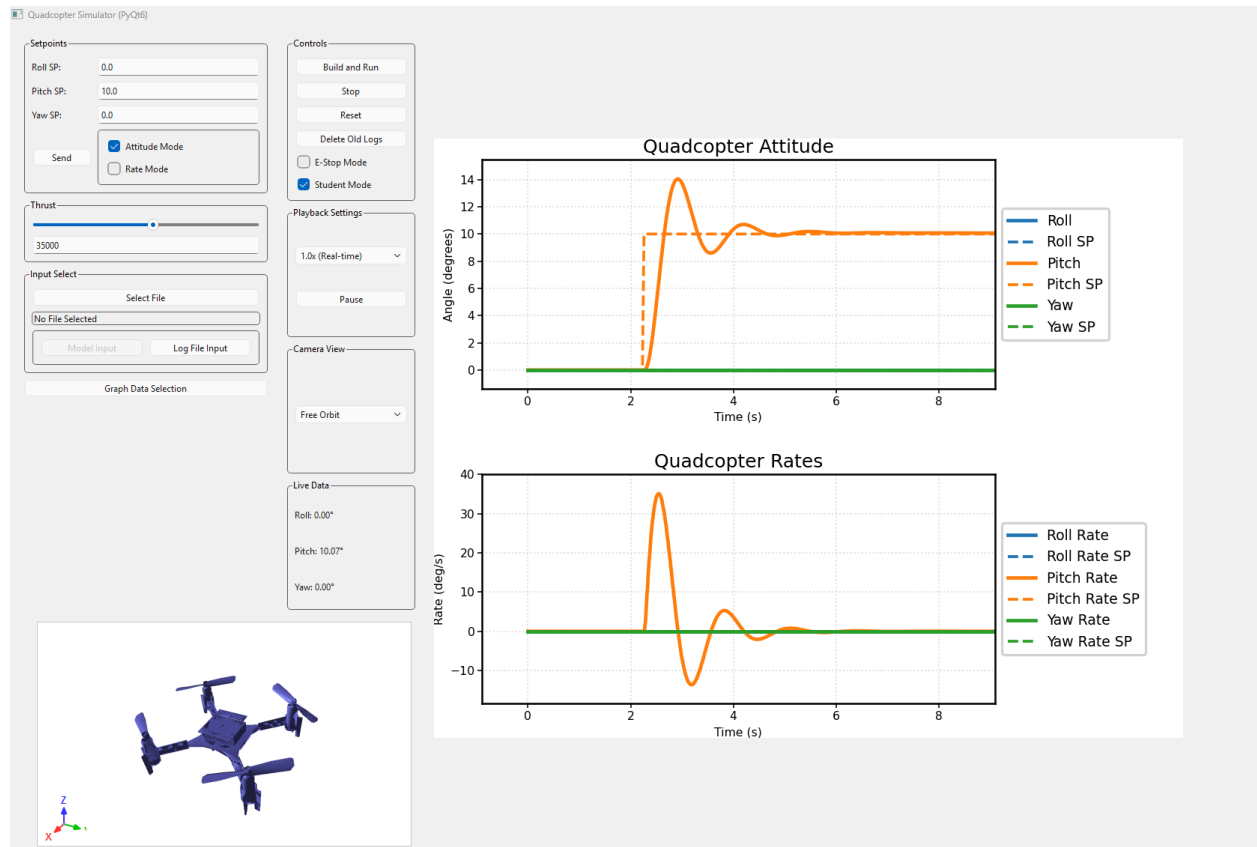
Roll: 10.07°

Pitch: 0.00°

Yaw: 0.00°







Part 2 - Testing Parameters

Note Thrust values: 16000-22000

Rate	Kp	Ki	Kd
Roll	250	0	15
Pitch	300	100	15
Yaw	500	200	5

Attitude	Kp	Ki	Kd
Roll	8	8	0
Pitch	8	8	0
Yaw	6	1	0

Student Effort

The following represents the percentage of effort from each team member for MP-4:

Cameron Gilbertson: 25%
Matthew Smith: 25%
Jessica Reiff: 25%
Collin Van Weelden: 25%